

12/3 Mat. analiz.

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УБТ 2.2.

6.1.3

$$f(x) = \frac{x^2 + 4}{x^3 + 1}$$

$$\square D(f): \begin{matrix} x^3 + 1 \neq 0 \\ x^3 \neq -1 \\ x \neq -1 \end{matrix}$$

$$\Rightarrow D(f) = (-\infty; -1) \cup (-1; +\infty)$$

6.1.4

$$f(x) = \sin \frac{1}{|x| - 2}$$

$$\square D(\sin x) = (-\infty; +\infty)$$

$$D_1(|x|) = (-\infty; +\infty)$$

$$D(f) = (-\infty; -2) \cup (-2; 2) \cup (2; +\infty)$$

$$D_2\left(\frac{1}{|x| - 2}\right): \begin{matrix} |x| - 2 \neq 0 \\ |x| \neq 2 \\ x \neq \pm 2 \end{matrix}$$

6.1.5

$$f(x) = \log_3(-x)$$

$$\square D(f): \begin{matrix} -x > 0 \\ x < 0 \end{matrix}$$

$$\Rightarrow D(f) = (-\infty; 0)$$

6.1.6

$$f(x) = \sqrt{x^2 - 7x + 10}$$

$$\square D(f): \begin{matrix} x^2 - 7x + 10 \geq 0 \\ (x-2)(x-5) \geq 0 \\ \begin{matrix} \text{минимум} & & \text{максимум} \\ 2 & & 5 \end{matrix} \end{matrix}$$

$$\begin{matrix} x^2 - 7x + 10 = 0 \\ D = 49 - 40 = 9 \\ x_{1,2} = \frac{7 \pm 3}{2} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \end{matrix}$$

$$\Rightarrow D(f) = (-\infty; 2] \cup [5; +\infty)$$

6.1.7

$$f(x) = x^2 + \operatorname{tg} x$$

□

$$D_1(f(x^2)) = (-\infty, +\infty)$$

$$D_2(\operatorname{tg} x) = \{x \mid x \neq \frac{\pi}{2} + k\pi\}, k \in \mathbb{Z}$$

$$D(f) = D_1 \cap D_2 = \{x \mid x \neq \frac{\pi}{2} + k\pi\}, k \in \mathbb{Z}$$



6.1.8

$$f(x) = \sqrt{x-7} + \sqrt{10-x}$$

□

$$D_1(\sqrt{x-7}) : \begin{cases} x-7 \geq 0 \\ x \geq 0 \end{cases}$$

$$\Rightarrow D_1(\sqrt{x-7}) = [7; +\infty)$$

$$D_2(\sqrt{10-x}) : \begin{cases} 10-x \geq 0 \\ x \leq 10 \end{cases}$$

$$\Rightarrow D_2(\sqrt{10-x}) = (-\infty; 10]$$

$$D(f) = D_1 \cap D_2 = [7; 10]$$



6.1.9

$$f(x) = \frac{\ln x}{\sqrt{|x^2-2|}}$$

□

$$D_1(\ln x) : x > 0 \Rightarrow D_1(\ln x) = (0; +\infty)$$

$$D_2(\sqrt{|x^2-2|}) : \begin{cases} |x^2-2| \geq 0 \\ x \in \mathbb{R} \end{cases}$$

$$\Rightarrow D_2(\sqrt{|x^2-2|}) = (-\infty; +\infty)$$

$$D_3\left(\frac{1}{\sqrt{|x^2-2|}}\right) : \sqrt{|x^2-2|} \neq 0$$

$$|x^2-2| \neq 0$$

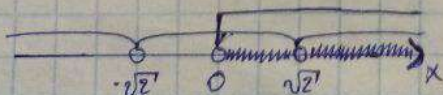
$$x^2-2 \neq 0$$

$$(x+\sqrt{2})(x-\sqrt{2}) = 0$$

$$x \neq \pm\sqrt{2}$$

$$\Rightarrow D_3\left(\frac{1}{\sqrt{|x^2-2|}}\right) = (-\infty; -\sqrt{2}) \cup (-\sqrt{2}; \sqrt{2}) \cup (\sqrt{2}; +\infty)$$

$$D(f) = D_1 \cap D_3 = (0; \sqrt{2}) \cup (\sqrt{2}; +\infty)$$



6.1.10

$$f(x) = \sqrt[3]{x+2} + \frac{1}{\sqrt{1-x}}$$

$$\square D_1(\sqrt[3]{x+2}) : \begin{array}{l} x+2 \geq 0 \\ x \geq -2 \end{array} \Rightarrow D_1(\sqrt[3]{x+2}) = [-2; +\infty)$$

$$D_2\left(\frac{1}{\sqrt{1-x}}\right) : \begin{array}{l} \sqrt{1-x} > 0 \\ 1-x \neq 0 \\ x \neq 1 \end{array} \quad \begin{array}{l} 1-x \geq 0 \\ x \leq 1 \end{array} \Rightarrow D_2\left(\frac{1}{\sqrt{1-x}}\right) = (-\infty; 1)$$

$$\underline{D(f)} = D_1 \cap D_2 = [-2; 1)$$



6.1.12

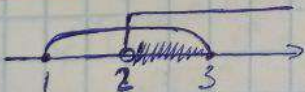
$$f(x) = \arccos(x-2) - \ln(x-2)$$

$$\square D(\arccos x) = [-1; 1] \Rightarrow \begin{array}{l} -1 \leq x-2 \leq 1 \\ 1 \leq x \leq 3 \end{array} \quad | +2$$

$$\Rightarrow D_1(\arccos(x-2)) = [1; 3]$$

$$D_2(\ln(x-2)) : \begin{array}{l} x-2 > 0 \\ x > 2 \end{array} \Rightarrow D_2(\ln(x-2)) = (2; +\infty)$$

$$\underline{D(f)} = D_1 \cap D_2 = (2; 3]$$



6.1.14

$$f(x) = x^2 - 8x + 20$$

$$\square f(x) = x^2 - 8x + 20 = (x^2 - 8x + 16) + 4 = (x-4)^2 + 4$$

$$(x-4)^2 \geq 0 \Rightarrow f(x) \geq 4$$

$$\Rightarrow \underline{E(f)} = [4; +\infty)$$



6.1.15

$$f(x) = 3^{-x^2}$$

$$\square E(x^2) = [0; +\infty)$$

$$3^{-x^2} = \frac{1}{3^{x^2}}$$

$$1) E(3^{x^2}) = [1; +\infty)$$

$$2) E\left(\frac{1}{3^{x^2}}\right) = \left[\frac{1}{+\infty}; \frac{1}{1}\right) = [0; 1)$$

$$\Rightarrow E(f) = (0; 1]$$



6.1.16

$$f(x) = 2 \sin x - 7$$

$$\square E(\sin x) = [-1; 1]$$

$$E(2 \sin x) = [-2; 2]$$

$$E(2 \sin x - 7) = [-2-7; 2-7] = [-9; -5]$$

$$\Rightarrow E(f) = [-9; -5]$$



6.1.17

$$f(x) = \frac{1}{x} + 4$$

$$\square E\left(\frac{1}{x}\right) = (-\infty; 0) \cup (0; +\infty)$$

$$E\left(\frac{1}{x} + 4\right) = (-\infty; 0+4) \cup (0+4; +\infty) = (-\infty; 4) \cup (4; +\infty)$$

$$E(f) = (-\infty; 4) \cup (4; +\infty)$$



6.1.20

$$f(x) = \frac{x+3}{x^2-1}$$

$$\square 1) f(0) = \frac{0+3}{0-1} = \frac{3}{-1} = -3$$

$$2) f(-2) = \frac{-2+3}{(-2)^2-1} = \frac{1}{3}$$

$$3) f(\sqrt{2}) = \frac{\sqrt{2}+3}{(\sqrt{2})^2-1} = \frac{\sqrt{2}+3}{1} = \sqrt{2}+3$$

$$4) f(-x) = \frac{-x+3}{(-x)^2-1} = \frac{3-x}{x^2-1}$$

$$5) f\left(\frac{1}{x}\right) = \frac{\frac{1}{x^2} + 3}{\left(\frac{1}{x}\right)^2 - 1} = \frac{\frac{1}{x^2} + 3}{\frac{1}{x^2} - 1}$$

$$6) f(a+1) = \frac{a+1+3}{(a+1)^2-1} = \frac{a+4}{a^2+2a+1-1} = \frac{a+4}{a^2+2a}$$

$$7) f(a)+1 = \frac{a+3}{a^2-1} + 1 = \frac{a^2-1+a+3}{a^2-1} = \frac{a^2+a+2}{a^2-1}$$

$$8) f(2x) = \frac{2x+3}{(2x)^2-1} = \frac{2x+3}{4x^2-1}$$



6.1.21.

$$f(x) = x^3 \cdot 2^x$$



$$1) f(1) = 1^3 \cdot 2^1 = 1 \cdot 2 = 2$$

$$2) f(-3) = (-3)^3 \cdot 2^{(-3)} = -27 \cdot \frac{1}{8} = -\frac{27}{8}$$

$$3) f(-\sqrt[3]{5}) = (-\sqrt[3]{5})^3 \cdot 2^{(-\sqrt[3]{5})} = \frac{-5}{2^{\sqrt[3]{5}}} = -\frac{5}{2^{\sqrt[3]{5}}}$$

$$4) f(-x) = (-x)^3 \cdot 2^{(-x)} = \frac{(-x)^3}{2^x} = -\frac{x^3}{2^x}$$

$$5) f(3x) = (3x)^3 \cdot 2^{3x} = 27x^3 \cdot 8^x$$

$$6) f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^3 \cdot 2^{\frac{1}{x}} = \frac{2^{\frac{1}{x}}}{x^3}$$

$$7) f\left(\frac{1}{x}\right) = \frac{1}{x^3 \cdot 2^x} = (x^3 \cdot 2^x)^{-1}$$

$$8) f(6-2) = (6-2)^3 \cdot 2^{6-2} = (6^3 - 6 \cdot 6^2 + 12 \cdot 6 - 8) \cdot 2^{6-2}$$



6.1.22.

$$\varphi(t) = \frac{\sqrt{t+5}}{t^2}$$



$$1) \varphi(-1) = \frac{\sqrt{-1+5}}{(-1)^2} = \frac{2}{1} = 2$$

$$2) \varphi(-5) = \frac{\sqrt{-5+5}}{(-5)^2} = \frac{\sqrt{0}}{25} = 0$$

$$3) \varphi\left(\frac{5}{4}\right) = \frac{\sqrt{\frac{5}{4}+5}}{\left(\frac{5}{4}\right)^2} = \frac{\sqrt{\frac{25}{4}}}{\frac{25}{16}} = \frac{5/2}{25/16} = \frac{8 \cdot 16^8}{8 \cdot 15_5} = \frac{8}{5} = 1,6$$

$$4) \varphi(z+3) = \frac{\sqrt{z+3+5}}{(z+3)^2} = \frac{\sqrt{z+8}}{z^2+6z+9}$$

$$5) \varphi(2t-1) = \frac{\sqrt{2t-1+5}}{(2t-1)^2} = \frac{\sqrt{2t+4}}{4t^2-4t+1}$$



6.1.24

$$1) f(x) = \frac{\sin x}{x}$$

□

$$D(f): x \neq 0 \Rightarrow D(f) = (-\infty, 0) \cup (0, +\infty)$$

$$f(-x) = \frac{\sin(-x)}{-x} = \frac{-\sin x}{-x} = \frac{\sin x}{x} = f(x)$$

$$\Rightarrow \underline{f(x) - \text{чет.}}$$



□

$$2) f(x) = x^5 + 3x^3 - x$$

$$D(f) = (-\infty; +\infty)$$

$$f(-x) = (-x)^5 + 3(-x)^3 - (-x) = -x^5 - 3x^3 + x = -(x^5 + 3x^3 - x) = -f(x)$$

$$\Rightarrow \underline{f(x) - \text{нечет.}}$$



□

$$3) f(x) = \sqrt{x}$$

$$D(f): x \geq 0 \Rightarrow D(f) = [0; +\infty)$$

$$f(-x) = \sqrt{-x} \neq \pm f(x) \Rightarrow$$

$$\Rightarrow \underline{f(x) - \text{обыч. фнк.}}$$



□

$$4) f(x) = \arcsin x$$

$$D(f) = [-1; 1]$$

$$f(-x) = \arcsin(-x) = -\arcsin x = -f(x) \Rightarrow$$

$$\Rightarrow \underline{f(x) - \text{нечет.}}$$



□

$$5) f(x) = \sin x + \cos x$$

$$D_1(\sin x) = (-\infty; +\infty)$$

$$D_2(\cos x) = (-\infty; +\infty)$$

$$D(f) = D_1 \cap D_2 = (-\infty; +\infty)$$

$$f(-x) = \sin(-x) + \cos(-x) = -\sin x - \cos x = -(\sin x + \cos x) = -f(x) \Rightarrow$$

$$\Rightarrow \underline{f(x) - \text{нечет.}}$$



□ 6) $f(x) = |x| - 2$

$D_1(|x|) = (-\infty; +\infty)$

$D(f) = (-\infty; +\infty)$

$f(-x) = |-x| - 2 = |x| - 2 = f(x) \Rightarrow$

$\Rightarrow \underline{f(x) = \text{чет}}$ 

7) $f(x) = \frac{3}{x^2 - 1}$

□ $D(f): \begin{matrix} x^2 - 1 \neq 0 \\ x^2 \neq 1 \\ x \neq \pm 1 \end{matrix}$

$\Rightarrow D(f) = (-\infty; -1) \cup (-1; 1) \cup (1; +\infty)$

$f(-x) = \frac{3}{(-x)^2 - 1} = \frac{3}{x^2 - 1} = f(x) \Rightarrow$

$\Rightarrow \underline{f(x) = \text{чет}}$ 

□ 8) $f(x) = x \cdot e^x$

$D(e^x) = (-\infty; +\infty)$

$D(f) = (-\infty; +\infty)$

$f(-x) = -x \cdot e^{-x} = -\frac{x}{e^x} \neq \pm f(x) \Rightarrow$

$\Rightarrow \underline{f(x) = \text{одн. вуг.}}$ 

6.1.26

$$1) f(x) = \cos \frac{x}{4}$$

$$\cos x: T = 2\pi$$

$$\text{тогда } \cos \frac{x}{4}: T = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$2) f(x) = |x|$$

при $x > 0$ $f(x)$ $\nearrow$ $\Rightarrow f(x)$ не период.
 при $x < 0$ $f(x)$ $\searrow$

$$\text{на } [0; +\infty) \Rightarrow \forall x \in (-\infty, +\infty) \exists T$$

$$3) f(x) = \operatorname{tg}(2x-1)$$

$$\operatorname{tg}(2x-1) = \operatorname{tg}(2x+\pi-1) = \operatorname{tg}(2(x+\frac{\pi}{2})-1)$$

$$\operatorname{tg}(2x-1): T = \frac{\pi}{2}$$

$$4) f(x) = \sin \frac{x}{2} + \operatorname{ctg} x$$

$$\sin \frac{x}{2}: T = \frac{2\pi}{2} = \pi$$

$$\operatorname{ctg} x: T = \pi$$

$$\sin(\frac{x}{2} + T_1) + \operatorname{ctg}(x + T_2)$$

$$\text{НОК}(T_1, T_2) = \text{НОК}(\pi, \pi) = \pi$$

$$5) f(x) = \sin 3x \cdot \cos 3x$$

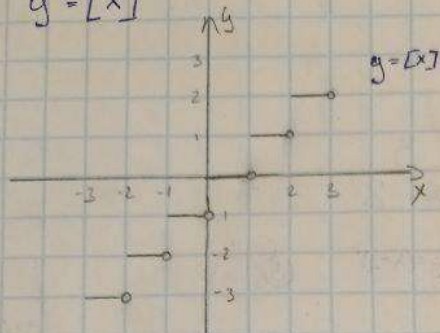
$$\sin 3x: T = \frac{2\pi}{3}$$

$$\cos 3x: T = \frac{2\pi}{3}$$

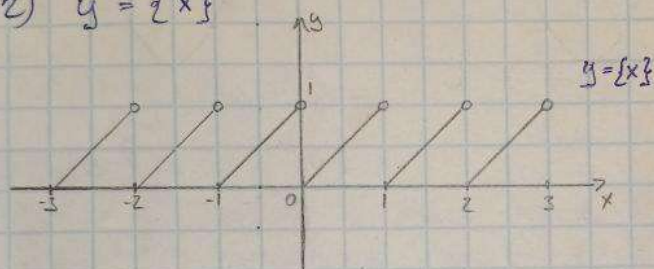
$$\text{НОК}(T_1, T_2) = \text{НОК}(\frac{2\pi}{3}, \frac{2\pi}{3}) = \frac{2\pi}{3}$$

6.1.27

1) $y = [x]$



2) $y = \{x\}$



6.1.29

$y = |x-3|$

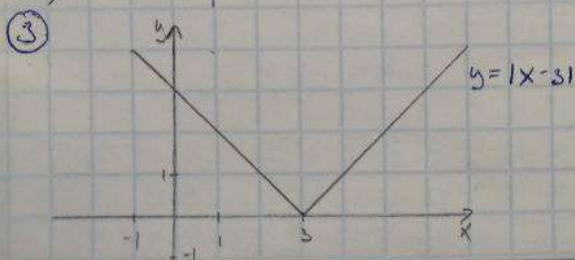
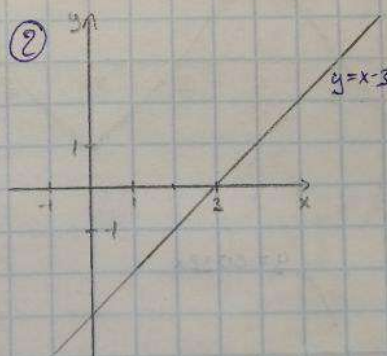
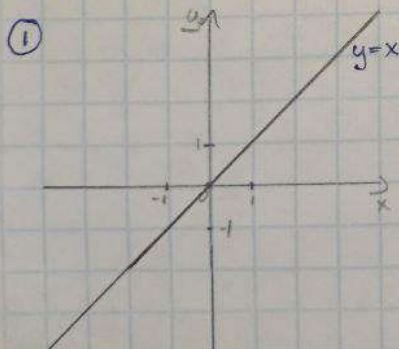
① $y = x$

② $y = x-3$

, сдвиг вправо на 3

③ $y = |x-3|$

, отраж. всех точек ниже O_x вверх, относ. O_x .



6.1.30

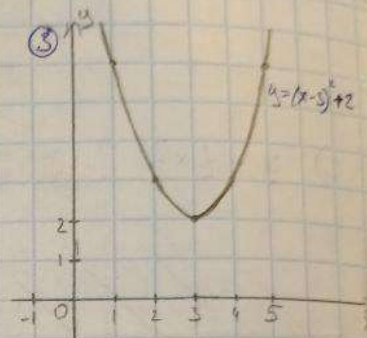
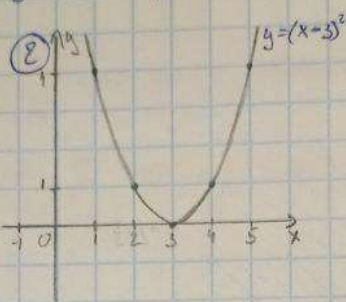
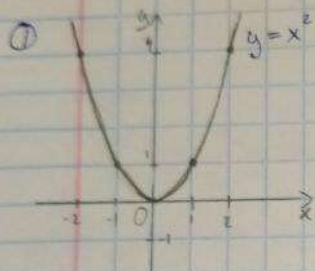
$$y = x^2 - 6x + 11$$

$$x^2 - 6x + 11 = x^2 - 6x + 9 - 9 + 11 = (x-3)^2 + 2$$

① $y = x^2$

② $y = (x-3)^2$, сдвиг вправо на 3

③ $y = (x-3)^2 + 2$, сдвиг вверх на 2



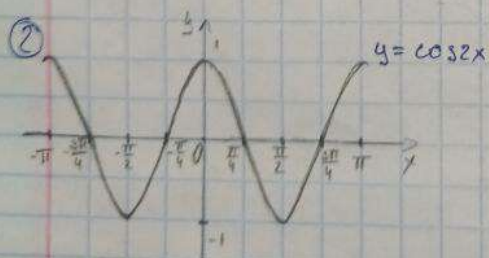
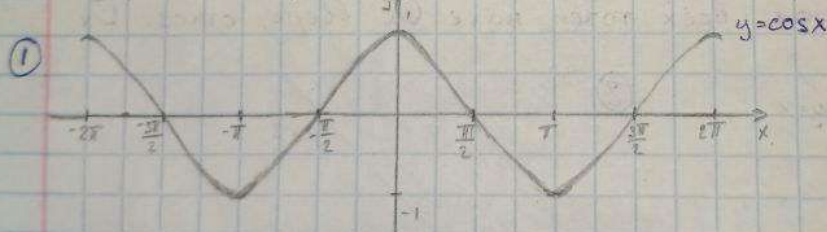
6.1.31

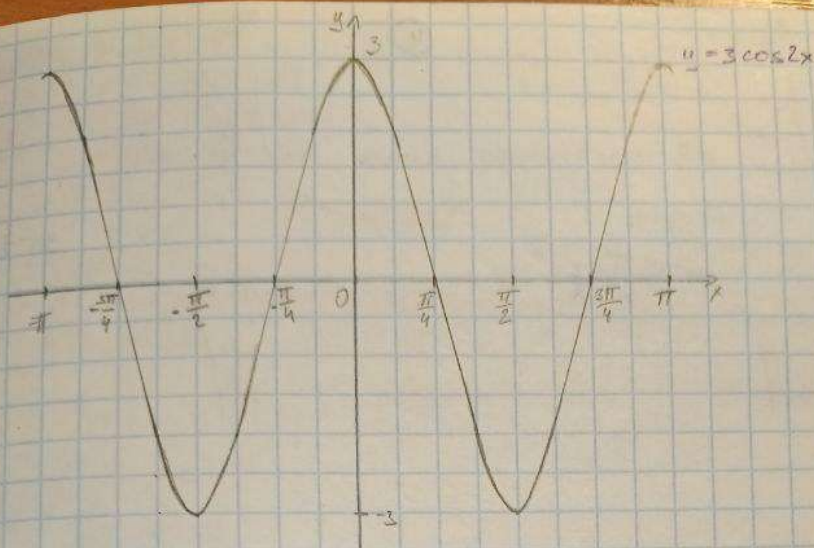
$$y = 3 \cdot \cos 2x$$

① $y = \cos x$

② $y = \cos 2x$, сжатие в 2 раза вдоль Ox

③ $y = 3 \cos 2x$, растяж. в 3 раза вдоль Oy



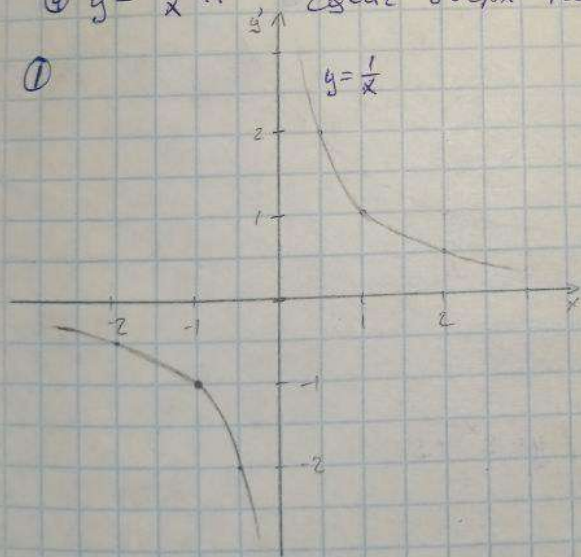


6.1.52

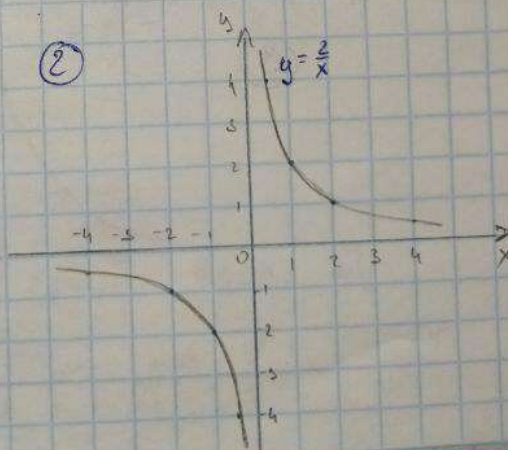
$$y = -\frac{2}{x} + 1$$

- ① $y = \frac{1}{x}$
- ② $y = \frac{2}{x}$, растя в 2 раза вдоль Oy
- ③ $y = -\frac{2}{x}$, отражение относительно Ox
- ④ $y = -\frac{2}{x} + 1$, сдвиг вверх на 1.

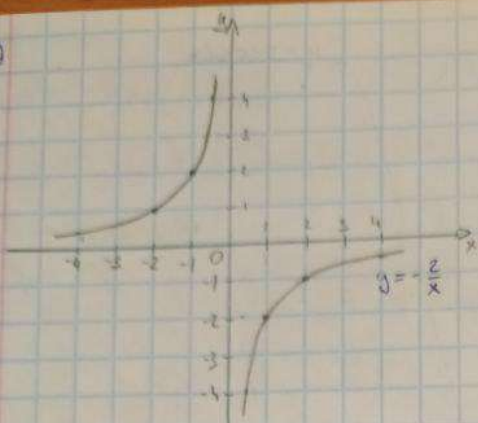
①



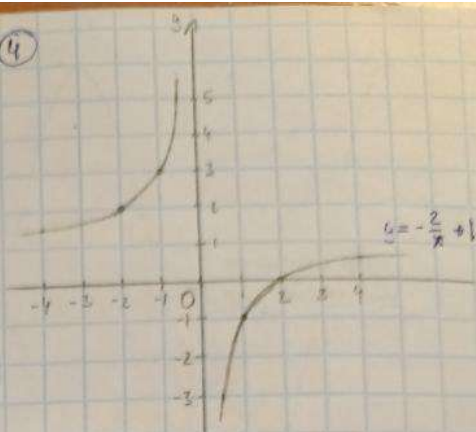
②



3



4



1.53.

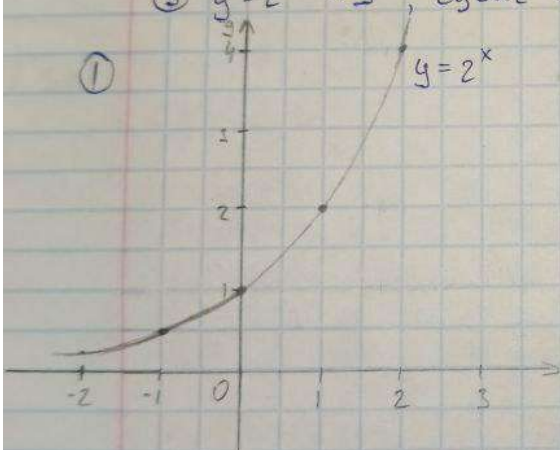
$$y = 2^{x-1} + 3$$

① $y = 2^x$

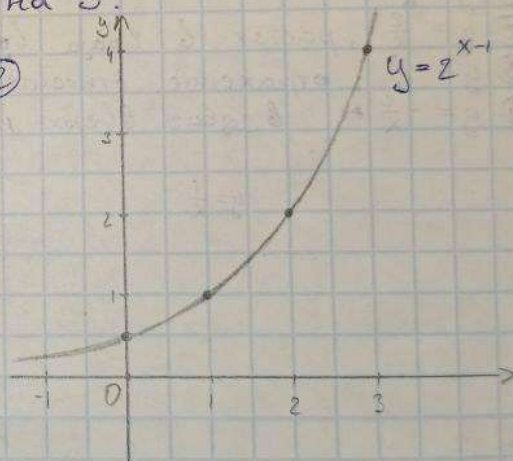
② $y = 2^{x-1}$, сдвиг вправо на 1

③ $y = 2^{x-1} + 3$, сдвиг вверх на 3.

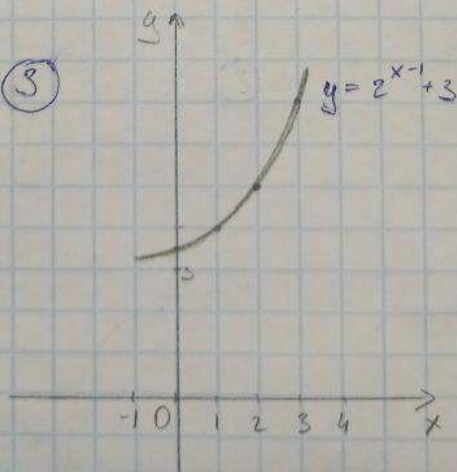
①



②



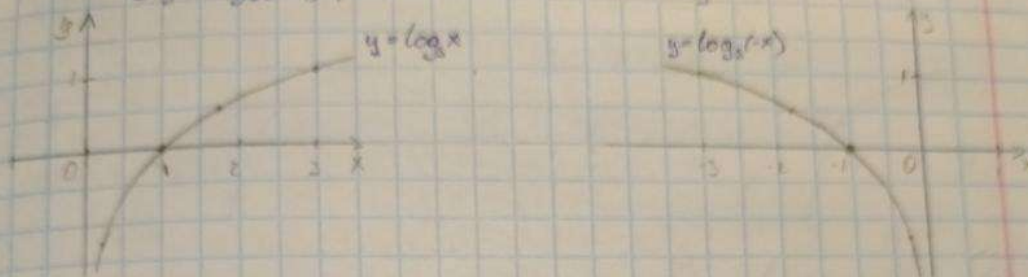
③



6.1.34

$$y = \log_3(-x)$$

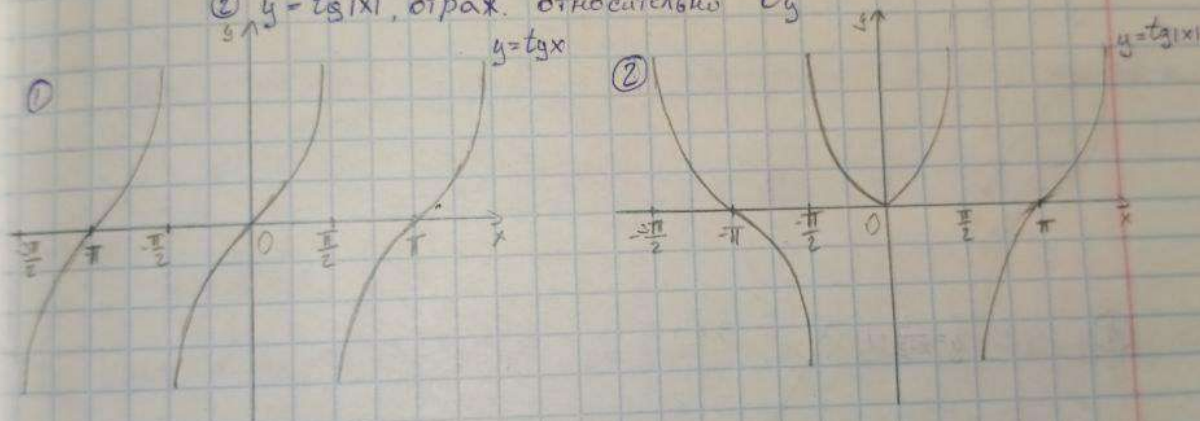
$$\textcircled{1} y = \log_3(x)$$

$$\textcircled{2} y = \log_3(-x), \text{отраж. относительно } O_y$$


6.1.35

$$y = \operatorname{tg} |x|$$

$$\textcircled{1} y = \operatorname{tg} x$$

$$\textcircled{2} y = \operatorname{tg} |x|, \text{отраж. относительно } O_y$$


6.1.36

$$y = \frac{x+4}{x+2}$$

$$\frac{x+4}{x+2} = \frac{x+2+2}{x+2} = 1 + \frac{2}{x+2}$$

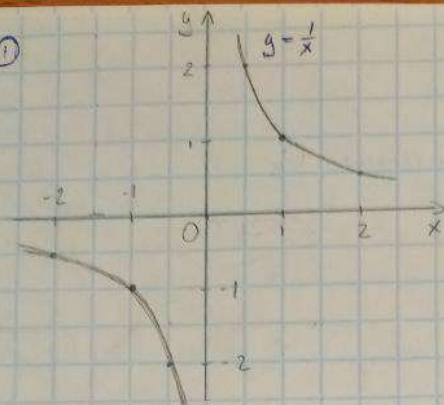
$$\textcircled{1} y = \frac{1}{x}$$

$$\textcircled{2} y = \frac{1}{x+2}, \text{сдвиг влево на 2}$$

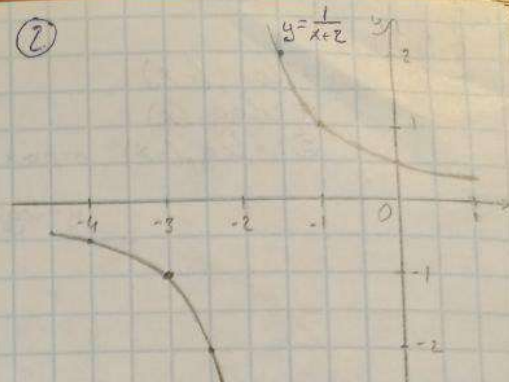
$$\textcircled{3} y = \frac{2}{x+2}, \text{растяж. в 2 раза вдоль } O_y$$

$$\textcircled{4} y = \frac{2}{x+2} + 1, \text{сдвиг вверх на 1}$$

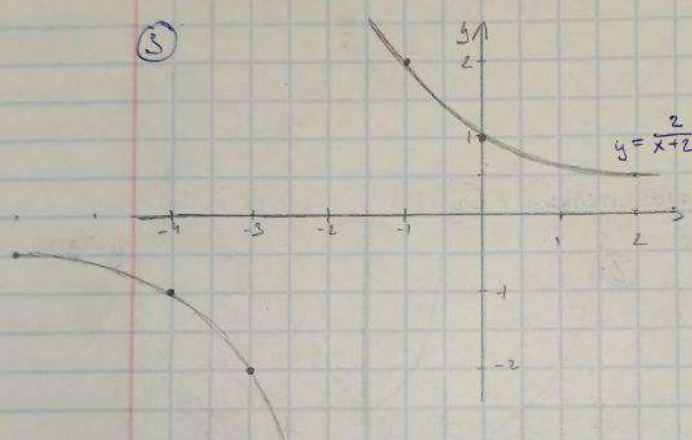
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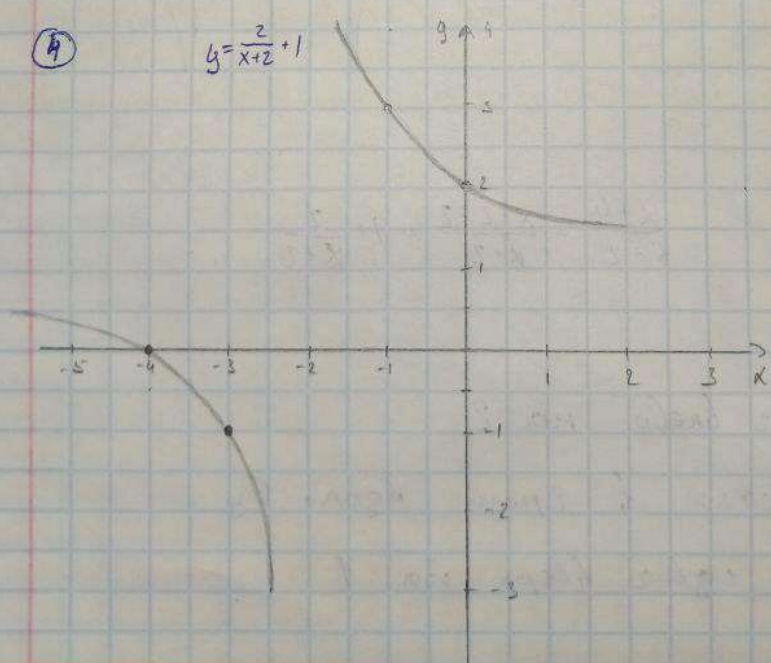
②



③



④



6.1.38

$$1) f(x) = e^x \quad g(x) = \ln x$$

$$f \circ g = f(g(x)) = e^{\ln x} = x$$

$$g \circ f = g(f(x)) = \ln(e^x) = x$$

$$2) f(x) = 3x + 1 \quad g(x) = 2x - 5$$

$$f \circ g = f(g(x)) = 3(2x - 5) + 1 = 6x - 15 + 1 = 6x - 14$$

$$g \circ f = g(f(x)) = 2(3x + 1) - 5 = 6x + 2 - 5 = 6x - 3$$

$$3) f(x) = |x| \quad g(x) = \cos x$$

$$f \circ g = f(g(x)) = |\cos x|$$

$$g \circ f = g(f(x)) = \cos |x|$$

6.1.41.

$$y = 3x + 5$$



на $(-\infty; +\infty)$ $f(x) \nearrow$, т.е. $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \Rightarrow \exists$ обрат. ф-я

$$x = \frac{y-5}{3} \Rightarrow \underline{y' = \frac{x-5}{3}}$$



6.1.42.

$$y = x^3 - 2$$



на $(-\infty; +\infty)$ $f(x) \nearrow$, т.е. $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \Rightarrow \exists$ обрат. ф-я

$$x = \sqrt[3]{y+2} \Rightarrow \underline{y' = \sqrt[3]{x+2}}$$



6.1.43

$$y = |x|$$



на $(-\infty; 0]$ $f(x) \searrow$; на $(0; +\infty)$ $\nearrow$

$$\forall y_0 > 0 \quad y_0 = |x| \Rightarrow \begin{matrix} x_1 = -x \\ x_2 = x \end{matrix} \Rightarrow$$

$\Rightarrow \nexists y'$, т.к. $\cap$ образ в 2-х точках



6.1.44

$$y = \frac{x-2}{x}$$

□

$$D(f): x \in \mathbb{R} \setminus \{0\}$$

на $(-\infty, +\infty)$ $f(x) \nearrow$, т.е. $x_1 < x_2 \Rightarrow f(x_1) < f(x_2) \Rightarrow \exists f'$

$$y = \frac{x-2}{x}; \quad yx = x-2, \quad x-yx=2; \quad x(1-y)=2; \quad x = \frac{2}{1-y} \Rightarrow$$

$$\Rightarrow \underline{y^{-1} = \frac{2}{1-y}}$$



6.1.45

$$f(x) = c$$

□

f -ция монотонная ($f(x)$ не $\nearrow$ и не $\searrow$)
 f -ция ограниченная



6.1.46

$$f(x) = \sin^2 x$$

□

f -ция ограниченная

$$[-1; 1]$$



6.1.47

$$f(x) = \operatorname{arctg} x$$

□

Строго монотонная ($f(x) \nearrow$)
 f -ция ограниченная
 $[-\frac{\pi}{2}; \frac{\pi}{2}]$



6.1.48

$$f(x) = -x^2 + 2x$$

□

f -ция не явл. ни монотонной, ни строго монотонной,
 ни ограниченной

на $(-\infty; 1)$ $f(x) \nearrow$; на $[1; +\infty)$ $\searrow$



6.1.43

$$f(x) = \frac{x+2}{x+5}$$



г-ция строго монотонная
на $(-\infty; +\infty)$ $f(x)$ ↗

6.1.54

$$\square 1) \frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$\frac{y^2}{4} = \frac{x^2}{9} - 1 \quad | \cdot 4$$

$$y^2 = \frac{4x^2}{9} - 4$$

$$y = \pm \sqrt{\frac{4x^2}{9} - 4}$$

$$y = \pm \sqrt{\frac{4x^2 - 36}{9}}$$

$$y = \pm \sqrt{\frac{4(x^2 - 9)}{9}}$$

$$y = \pm \frac{2\sqrt{x^2 - 9}}{3}$$

$$2) x + |y| = 1$$

$$|y| = 1 - x$$

$$y = \pm(1-x) \quad \text{при } x \leq 1$$

$$3) e^y - \sin y = x^2$$

6.1.55

$$\square y + \cos y - x = 0$$

$$1) A(1; 0)$$

$$0 + \cos 0 - 1 = 0$$

$$0 = 0$$

⇒ принадлежит.

$$2) B(0; 0)$$

$$0 + \cos 0 - 0 = 1$$

$$1 \neq 0$$

⇒ не принадлежит.

$$3) C\left(\frac{\pi}{2}; \frac{\pi}{2}\right)$$

$$\frac{\pi}{2} + \cos \frac{\pi}{2} - \frac{\pi}{2} = 0$$

$$0 = 0$$

⇒ принадлежит.

$$4) D(\pi-1; \pi)$$

$$\pi + \cos \pi - \pi + 1 = 0$$

$$0 = 0$$

⇒ принадлежит.